



UNIVERSITÀ DEGLI STUDI DI PADOVA

Lab 1: 4th-order elliptic low-pass filter

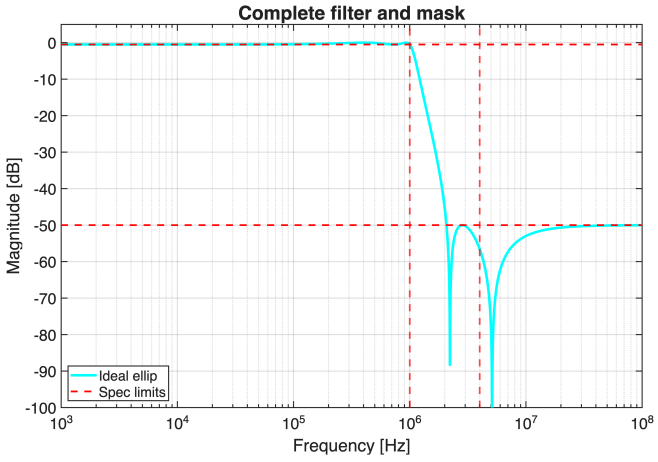
Integrated Circuits for Signal Processing

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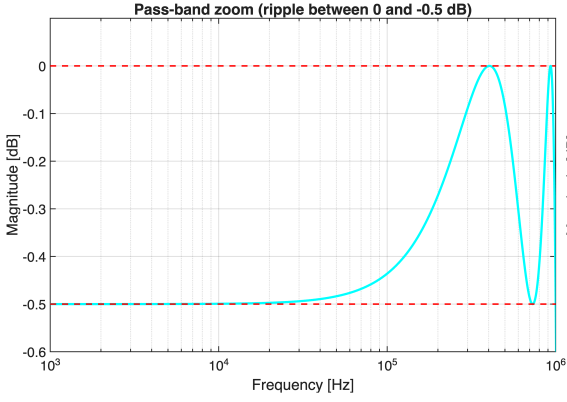
2 November 2026

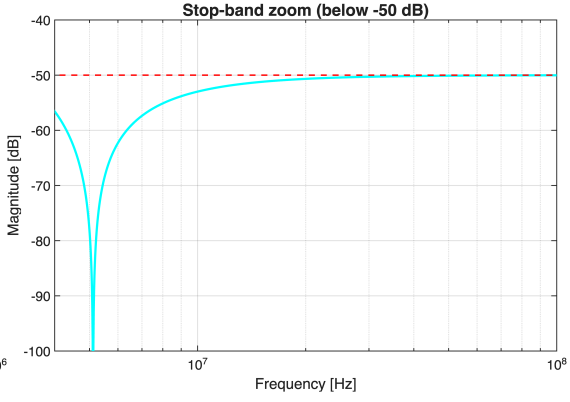
Matlab Mathematical Result & Specifications

- 4th-order low-pass elliptic filter, ellipord gives minimumm order $N = 3$
- Pass band edge $f_p = 1$ MHz, stop pand edge $f_s = 4$ MHz
- In-band ripple ≤ 0.5 dB, stop-band attenuation ≥ 50 dB
- Output noise $< 120\ \mu V_{\mathrm{rms}}$



Pass Band & Stop Band Zoom





Biquad Decomposition

- The overall filter transfer function can be decomposed into two separate second-order transfer functions.
- For each transfer function, we determine k_0, k_1, k_2, Q, and ω_0. Biquad 1 has the lower Q and comes first in the cascade, while Biquad 2 has higher Q and all the DC gain of the total transfer function

$$H(s) = \frac{k_2 s^2 + k_1 s + k_0}{\frac{s^2}{\omega_0^2} + \frac{s}{Q\omega_0} + 1}$$

Biquad 1

Parameter	Value
ω_0 [rad/s]	4.0039×10^6
Q	7.2264×10^{-1}
k_0	1.0000
k_1	0.0000
k_2	9.6813×10^{-16}

Biquad 2

Parameter	Value
ω_0 [rad/s]	6.4771×10^6
Q	3.3831
k_0	9.4406×10^{-1}
k_1	0.0000
k_2	4.8566×10^{-15}

Component Sizing from the Noise Specification

- The extracted parameters constrain the RC time constants. Component values must also meet the noise limit; deriving each noise-source-to-output transfer function is a long procedure, so we use a simpler estimate.
- Output noise power scales as $16kT/C$. With equal capacitors, a noise check lets us rescale all capacitances proportionally to the noise limit of the specifications.

$$C_a = C_b = 16 \cdot \frac{kT}{V^2_{\mathrm{noise,RMS}}} \simeq 4.6\,\mathrm{pF}$$

Biquad 1

Component	Initial	After noise check
$C_a = C_b$	4.6 pF	9.2 pF
R_1	39.2358 kΩ	19.6179 kΩ
R_3	54.2948 kΩ	27.1474 kΩ
R_4	54.2948 kΩ	27.1474 kΩ
R_5	−54.2948 kΩ	−27.1474 kΩ
C_4	−0.0714 pF	−0.1428 pF

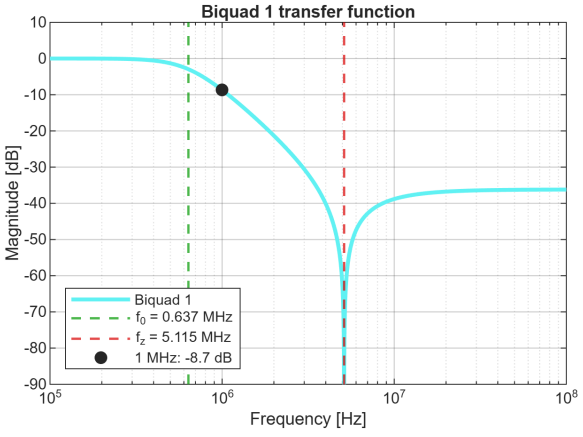
Biquad 2

Component	Initial	After noise check
$C_a = C_b$	4.6 pF	9.2 pF
R_1	113.5463 kΩ	56.7732 kΩ
R_3	33.5629 kΩ	16.7814 kΩ
R_4	35.5516 kΩ	17.7758 kΩ
R_5	−33.5629 kΩ	−16.7814 kΩ
C_4	−0.9372 pF	−1.8745 pF

First biquad: transfer function (MATLAB)

Parameters			
f_0	Q	f_z	k_0
0.637 MHz	0.72	5.115 MHz	1.00

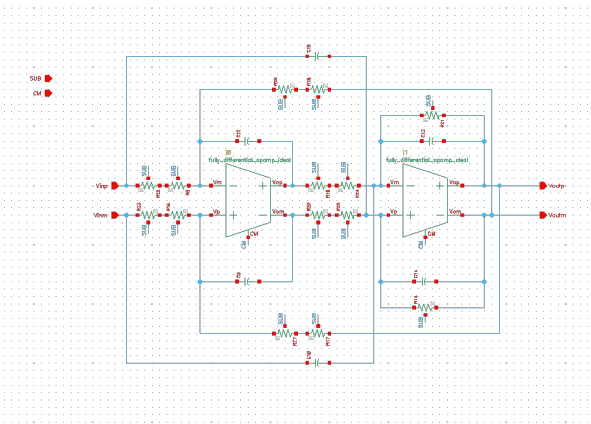
- Low Q: no peaking, flat at 0 dB at low frequency
- Zero at 5.1 MHz: deep notch, then -36 dB at high frequency
- It attenuates the upper part of the pass band: -8.7 dB at 1 MHz, recovered by biquad 2



First biquad: circuit in Virtuoso

Circuit

- Two fully differential integrators with the ideal amplifier model
- MIM capacitors and RNHR1000 poly resistors (umc13mmrf)

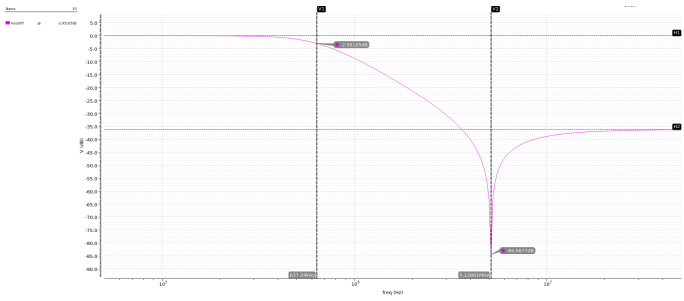


First biquad: Virtuoso, real components

Test bench

- Same amplifier model, with MIM capacitors and RNHR1000 poly resistors (umc13mmrf)

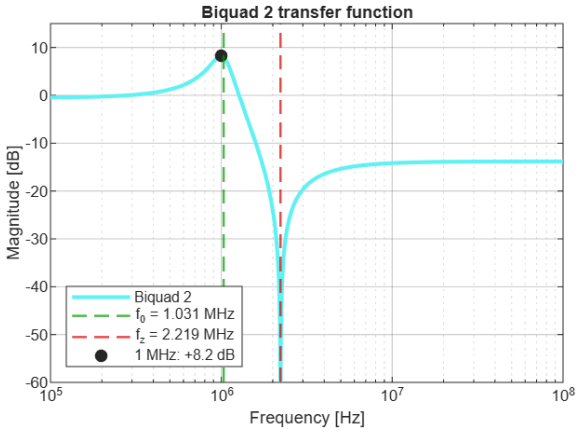
	MATLAB	Virtuoso
f_0 [MHz]	0.637	0.637
f_z [MHz]	5.115	5.129
DC gain [dB]	0.0	0.0
HF gain [dB]	-36.2	≈ -36.2
Gain at 1 MHz [dB]	-8.7	≈ -8.7



Second biquad: transfer function (MATLAB)

Parameters			
f_0	Q	f_z	k_0
1.031 MHz	3.38	2.219 MHz	0.944

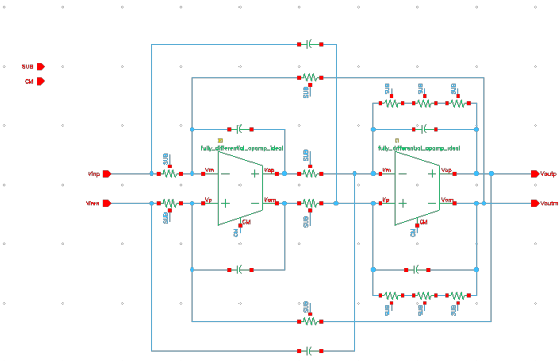
- High Q: peak of $+8.2$ dB at 1.0 MHz, right at the pass band edge
- It compensates the -8.7 dB of biquad 1 at 1 MHz: the cascade stays within -0.5 dB
- DC gain -0.5 dB ($k_0 = 0.944$) and zero at 2.2 MHz, the closest to the pass band



Second biquad: circuit in Virtuoso

Circuit

- Same structure as biquad 1, with the values of the second pair of poles and zeros
- MIM capacitors and RNHR1000 poly resistors (umc13mmrf)

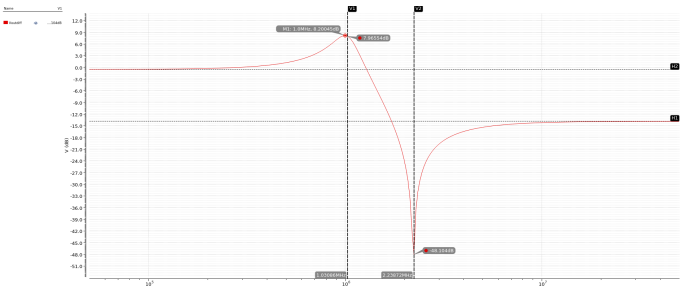


Second biquad: Virtuoso, real components

Test bench

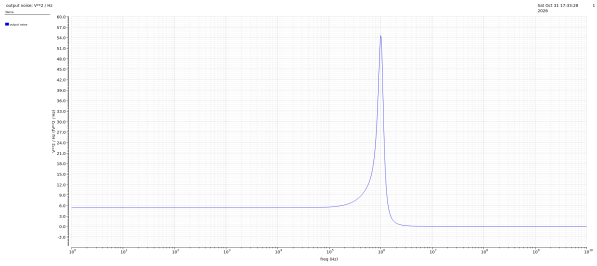
- Same amplifier model, with MIM capacitors and RNHR1000 poly resistors (umc13mmrf)

	MATLAB	Virtuoso
f_0 [MHz]	1.031	1.031
f_z [MHz]	2.219	2.239
DC gain [dB]	-0.5	≈ -0.5
HF gain [dB]	-13.8	≈ -13.7
Gain at 1 MHz [dB]	$+8.2$	$+8.2$

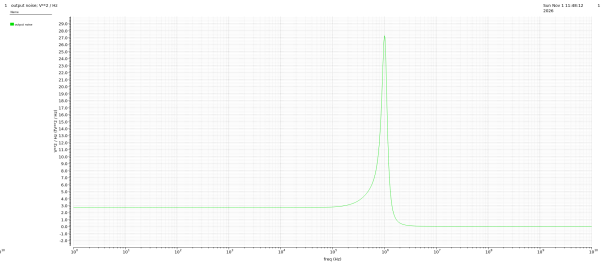


Power Spectral Density of Noise

$C = 4.6\,\mathrm{pF}$



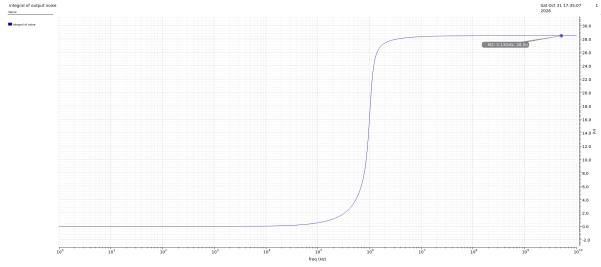
$C = 9.2\,\mathrm{pF}$



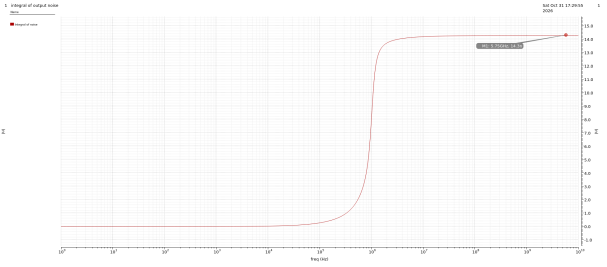
- The PSD sums resistor-noise contributions weighted by their transfer functions.
- Integrating from 1 Hz to 10 GHz gives the noise power of interest.

Integrated Noise Power

$C = 4.6\,\mathrm{pF}$



$C = 9.2\,\mathrm{pF}$

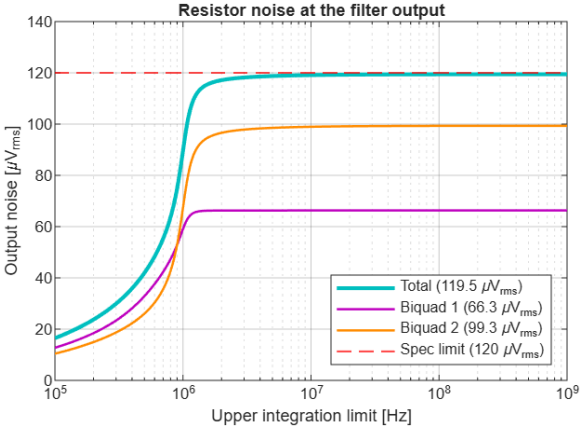


- RMS noise is the square root of the integrated noise power.
- Doubling the capacitance reduces the initial noise to the target; the final output is 119.4 $\mu\mathrm{V_{rms}}$.

Noise analysis (MATLAB)

- Thermal noise of each resistor, $4kTR$ ($T = 300$ K), $\times 2$ for the fully differential structure
- Referred to the filter output through the noise transfer function and the following biquad
- Integrated from 1 Hz to 10 GHz: total 119.5 μV_{rms}, below the 120 μV_{rms} limit
- Biquad 2 dominates (99.3 vs 66.3 μV_{rms}); the noise is collected in the pass band

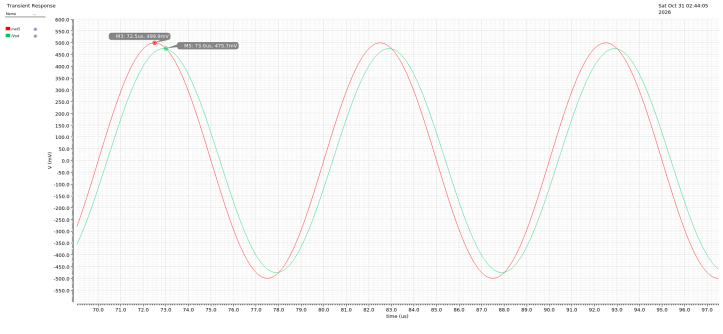
μV_{rms}	R_1	R_3	R_4	R_5	Tot.
Biquad 1	37.8	32.1	31.2	31.2	66.3
Biquad 2	30.0	55.2	53.6	55.2	99.3
Filter (rms sum)			119.5		



Transient Response 100 kHz

- The plot shows the input and output sinusoids together for a quick comparison.
- The transfer function introduces a small attenuation and a phase shift in the output sinusoid.
- The attenuation remains within the specification:

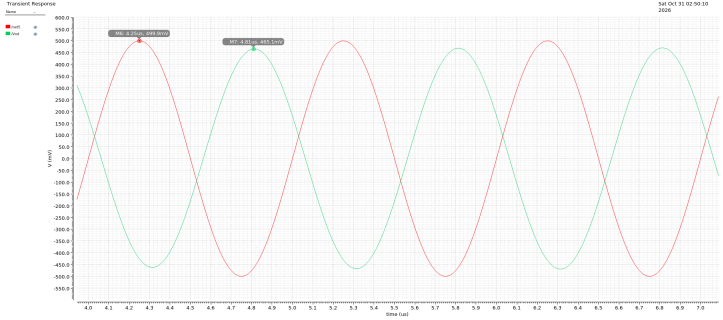
$$A_{1,\mathrm{dB}} = 20 \log_{10}\left(\frac{A_{\mathrm{out}}}{A_{\mathrm{in}}}\right) = -0.43\,\mathrm{dB}$$



Transient Response 1 MHz

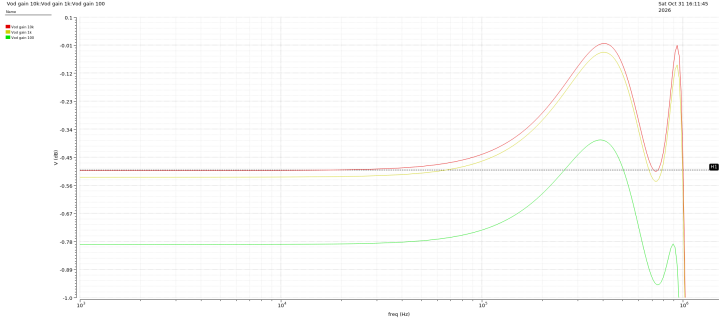
- At the bandwidth limit, the output sinusoid exhibits a large phase shift.
- The attenuation is slightly outside the specification:

$$A_{2,\mathrm{dB}} = 20\log_{10}\left(\frac{A_{\mathrm{out}}}{A_{\mathrm{in}}}\right) = -0.63\,\mathrm{dB}$$



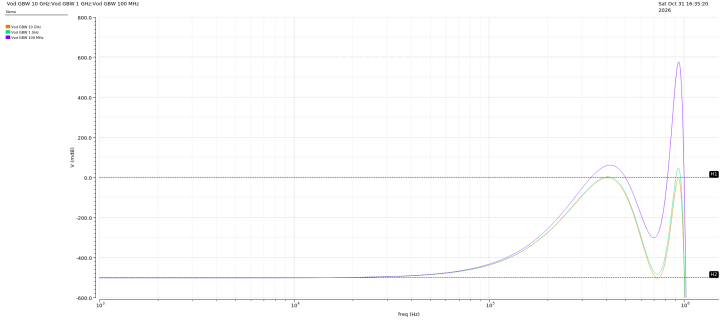
Limited OPAMP gain effects

- Reducing the gain causes a drop before the edge, pushing the response outside the specifications.
- Tested gains: 100, 1k, and 10k.



Limited OPAMP bandwidth effects

- Reducing the bandwidth creates a higher peak at the pass-band limit.
- Tested bandwidths: 100 MHz, 1 GHz, and 10 GHz.



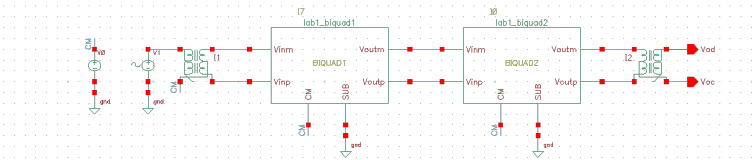




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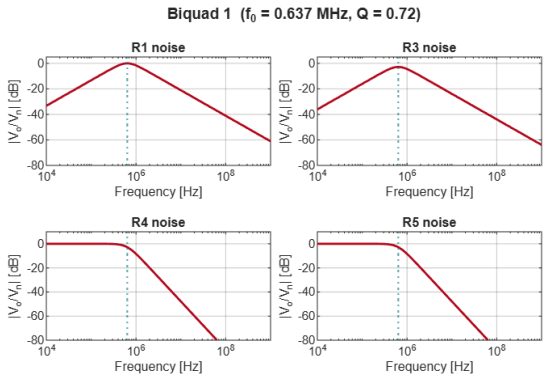
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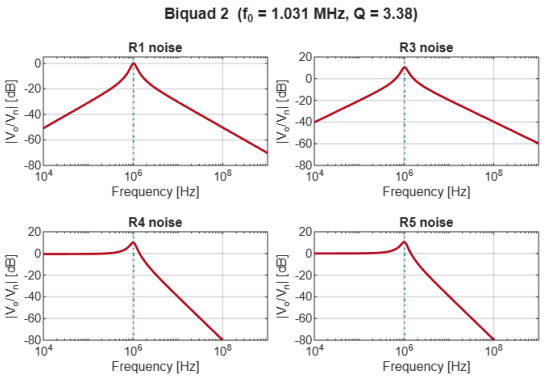
Backup: test bench of the complete filter



- Cascade of `lab1_biquad1` and `lab1_biquad2`, with common mode (CM) and substrate (SUB) connected
- Input: DC source at the common mode plus a sine source (500 mV, frequency `f1`)
- Balun at input and output: `Vod` is the differential output, `Voc` the common-mode output
- Common mode $= V_{DD}/2 = 0.5$ V ($V_{DD} = 1$ V): it biases the amplifier correctly (0 V would switch it off) and is the zero-signal level, so the maximum amplitude is $V_{DD}/2$

Backup: noise transfer functions of the two biquads





Biquad 1 ($Q = 0.72$): R_1, R_3 band-pass, R_4, R_5 low-pass

Biquad 2 ($Q = 3.38$): peaks of about $+10$ dB $\approx 20 \log Q$ on R_3, R_4, R_5